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//
// Secure Multi-User Chat Application in C++
// Features:
// • Multiple Users
// • Send / Receive Messages
// • Save Chats in File
// • Load Previous Chats
// • Screenshot Detection Simulation
// • Auto Delete Chats when Screenshot Detected
//
```

```
#include <iostream>
#include <fstream>
#include <vector>
#include <ctime>
#include <cstdio>
```

```
using namespace std;
```

```
struct Message {
    string sender;
    string receiver;
    string text;
    string time;
};
```

```
vector<Message> chats;
```

```
// Get current time
string currentTime() {
    time_t now = time(0);
    string dt = ctime(&now);
    dt.pop_back();
    return dt;
}
```

```
// Save chats to file
void saveChats() {
    ofstream file("chats.txt");

    for (auto &m : chats) {
        file << m.sender << "|"
            << m.receiver << "|"
            << m.text << "|"
            << m.time << endl;
    }
}
```

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file.close();
```

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}

// Load chats from file
void loadChats() {
    chats.clear();

    ifstream file("chats.txt");

    string sender, receiver, text, time;

    while (getline(file, sender, '|')) {
        getline(file, receiver, '|');
        getline(file, text, '|');
        getline(file, time);

        chats.push_back({sender, receiver, text, time});
    }

    file.close();
}

// Send message
void sendMessage(string sender) {
    Message msg;

    msg.sender = sender;

    cout << "\nEnter Receiver Name: ";
    cin >> msg.receiver;

    cin.ignore();

    cout << "Enter Message: ";
    getline(cin, msg.text);

    msg.time = currentTime();

    chats.push_back(msg);

    saveChats();

    cout << "\nMessage Sent Successfully!\n";
}

// View messages
void viewMessages(string user) {

    cout << "\n===== CHAT MESSAGES =====\n";

```

```

bool found = false;

for (auto &m : chats) {

    if (m.sender == user || m.receiver == user) {

        cout << "\nFrom : " << m.sender;
        cout << "\nTo   : " << m.receiver;
        cout << "\nText : " << m.text;
        cout << "\nTime : " << m.time << endl;

        cout << "-----\n";

        found = true;
    }
}

if (!found) {
    cout << "\nNo messages found.\n";
}
}

// Simulate screenshot detection
void screenshotDetected() {

    cout << "\n⚠ SCREENSHOT DETECTED!\n";
    cout << "Deleting all chat messages...\n";

    chats.clear();

    remove("chats.txt");

    cout << "All chats deleted successfully.\n";
}

int main() {

    loadChats();

    string currentUser;

    cout << "=====\n";
    cout << "  MULTI USER CHAT APPLICATION\n";
    cout << "=====\n";

    cout << "\nEnter Your Username: ";
    cin >> currentUser;

```

```
int choice;

while (true) {

    cout << "\n===== MENU =====\n";
    cout << "1. Send Message\n";
    cout << "2. View Messages\n";
    cout << "3. Screenshot Detected\n";
    cout << "4. Exit\n";

    cout << "\nEnter Choice: ";
    cin >> choice;

    switch (choice) {

        case 1:
            sendMessage(currentUser);
            break;

        case 2:
            viewMessages(currentUser);
            break;

        case 3:
            screenshotDetected();
            break;

        case 4:
            cout << "\nExiting Application...\n";
            return 0;

        default:
            cout << "\nInvalid Choice!\n";
    }
}

return 0;
}
```